

AI Value Sprint Canvas

Turn a business challenge into a small, measurable and responsible AI pilot.

TEAM / COOPERATIVE Deep Learning Team Project		CASE / AREA Viticulture / Vineyard inspection	DATE 13/09/2026
1 Business challenge Vineyard managers and agronomists need practical ways to identify potentially diseased vines from field observations so that limited inspection time can be focused where it is most useful.	2 Evidence from reality A real-world RGB image dataset from northern Portugal contains five labelled classes: Healthy, Downy Mildew, Powdery Mildew, Esca Complex and Erineum Mite. Images were collected under natural vineyard conditions.	3 People involved Primary users: vineyard managers and agronomists. A human expert remains responsible for inspection, follow-up and final judgement; the model is decision support, not automated diagnosis.	
4 Opportunity question How might we use vineyard photographs to prioritise which vines should receive human inspection and follow-up, while keeping the expert in control?	5 AI-enabled move Use a CNN image classifier to recognise visible symptom patterns and classify each photograph into one of the five target classes, supporting prioritisation of human inspection.	6 Data and sources Needed: labelled RGB grapevine-leaf photographs. Source: Grapevine Leaves RGB Images of Disease Symptoms (Portela et al., 2026). Use one consistent train/validation/test split; resize, normalise and encode five classes.	
7 Human workflow 1. AI classifies a vineyard photograph into one of five classes. 2. Human expert reviews the result and field context. 3. Expert decides whether inspection and follow-up are required.	8 Trust and controls Safeguards: human review; no automated diagnosis; reproducible train/validation/test split. Risks include natural-lighting variation, cultivar imbalance and symptom ambiguity.	9 Test Horizon First validate data/pipeline and model performance, then compare results against a baseline before a Go/No-Go decision.	
★ Value to create		Focus limited vineyard inspection time on vines most likely to need expert attention, while preserving human judgement.	
Decision → Evidence → AI move → Responsible action			